

AI ERP System -- Technology & Product Overview

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Executive Summary

The AI ERP System is a vertically integrated, AI-native enterprise resource planning platform purpose-built for CPG (Consumer Packaged Goods) companies, manufacturers, and brands managing complex supply chains with copackers, vendors, and multi-warehouse operations.

Unlike legacy ERP systems that bolt on AI as an afterthought, our platform was designed from inception around autonomous AI agents that actively manage supply chain operations, procurement, production planning, and financial workflows -- with configurable human-in-the-loop oversight for high-value decisions.

The system consolidates 10+ point solutions (inventory management, order management, accounting, CRM, procurement, manufacturing, logistics, HR, legal, and document management) into a single, modern platform, dramatically reducing software spend and integration complexity for mid-market CPG brands.

Product Scope & Modules

Core Operational Modules

Module	Key Capabilities
Sales & Order Management	Sales orders, customer management, status tracking (pending through delivered), Shopify sync, returns
Operations & Inventory	Multi-warehouse stock tracking, lot/expiration management, real-time levels, reservations, physical counts, inter-warehouse transfers
Manufacturing & BOM	Multi-level bill of materials with versioning, work orders, production batches, raw material tracking, yield analysis
Procurement	Purchase order lifecycle, vendor management, goods receiving, three-way match (PO/receipt/invoice), AI-suggested reorders
Logistics & Freight	Shipment tracking, freight RFQ management, carrier comparison, booking, customs clearance with 12+ document types
Finance & Accounting	Chart of accounts, double-entry journal entries, invoicing (including recurring), payments, account reconciliation, QuickBooks two-way sync
CRM & Fundraising	Contact pipeline management, investor tracking, campaign management for funding rounds, cap table modeling, communication logging
HR & Payroll	Employee profiles, department hierarchy, compensation tracking, payroll processing
Legal & Contracts	Contract lifecycle management (7 types), renewal/expiration tracking, dispute management, document storage
Projects & Tasks	Project management with budgets, task assignment, milestone tracking, time tracking

AI & Automation Layer

Capability	Description
AI Command Bar	Natural language interface (Cmd+K) for all operations -- create POs, record payments, generate invoices, transfer inventory, and more via plain English
Conversational AI Assistant	20+ integrated tools for data analysis, email generation, document parsing (including OCR), forecasting, and anomaly detection
Autonomous Supply Chain Orchestrator	14 automated workflows running on schedule/event/threshold triggers covering demand forecasting, production planning, procurement, fulfillment, and payment processing
AI Agent Scheduler	Rule-based automation engine for inventory reorders, PO generation, vendor follow-ups, payment reminders, quality checks
Smart Document Import	Vision-based OCR for scanned PDFs with automatic entity extraction, confidence scoring, and record creation
Email Intelligence	IMAP-based inbox scanning with AI categorization, auto-reply rules, and automatic task creation

Partner Portals

Portal	Access Model	Capabilities
Copacker Portal	Authenticated	Inventory reporting, shipment management, customs document upload
Vendor Portal	Authenticated	PO management, shipment tracking, document upload
Public Supplier Portal	Token-based (no login)	Export document upload, freight configuration, HS codes, incoterms
Investor Data Rooms	Share-link with optional password	Secure document sharing, visitor tracking, access logging

Integrations

Integration	Scope
QuickBooks Online	OAuth 2.0, two-way sync of customers, vendors, invoices, chart of accounts, payments
Shopify	Order import, customer sync, inventory sync, fulfillment updates
Google Workspace	Gmail, Sheets, Docs, Drive (full CRUD and sync)
SendGrid	Transactional email, template management, delivery tracking via webhooks
Fireflies.ai	Meeting transcription, action item extraction, automatic task creation
IMAP	Inbound email scanning from any mailbox
Airtable	Data import from Airtable bases

Technology Architecture

Stack

Layer	Technology
Frontend	React 19, TypeScript, Tailwind CSS 4, Radix UI, Wouter routing
Backend	Express.js, tRPC (end-to-end type-safe RPC), Node.js
Database	MySQL with Drizzle ORM
Build	Vite 7 (frontend), esbuild (backend), TypeScript 5.9
Testing	Vitest with 31 test suites
Auth	OAuth 2.0 (Google, Apple, Microsoft, GitHub) + email/password with PBKDF2 hashing, JWT sessions
AI/ML	LLM-powered agents, vision-based OCR, autonomous workflow engine
Infra	Deployable to Vercel, Railway, or any Node.js host; S3-compatible storage

Architecture Highlights

- **End-to-End Type Safety:** tRPC provides compile-time type checking from database schema through API layer to React components -- eliminating an entire class of runtime errors
- **Monorepo with Shared Types:** Single repository with `client/`, `server/`, and `shared/` directories; shared Zod schemas validate data at every boundary

- **169 Database Tables:** Comprehensive relational schema (5,076 lines) covering all business domains with full referential integrity
- **90 Page Components, 69 UI Components:** Rich, production-grade frontend built on Radix UI primitives with consistent design system
- **128,000+ Lines of TypeScript:** Fully typed codebase with zero JavaScript files
- **Autonomous Agent Architecture:** Three auto-starting background services (Supply Chain Orchestrator, AI Agent Scheduler, Email Queue Worker) that operate independently with configurable human oversight

Security & Access Control

- **10 Role Types:** admin, finance, procurement, ops, plant, legal, exec, copacker, vendor, contractor -- each with granular module-level permissions
 - **Granular Permission Overrides:** Admins can customize access per user beyond role defaults
 - **Secure Authentication:** PBKDF2 with 100,000 iterations, JWT in HTTP-only cookies, rate limiting
 - **Audit Trail:** All autonomous AI decisions are logged with full traceability
 - **Data Room Security:** Password-protected share links with visitor tracking and access logging
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Autonomous AI Capabilities -- Competitive Differentiator

The autonomous workflow engine is the core differentiator of this platform. Rather than requiring manual intervention for routine operations, the system runs 14 automated workflows:

Workflow Schedule

Workflow	Trigger	Approval Threshold
Daily Demand Forecasting	6 AM daily	None (advisory)
Production Planning	7 AM daily	None (advisory)
Material Requirements Planning	8 AM daily	> \$1,000
Procurement Processing	Event-driven	Configurable
Inventory Reorder Check	Threshold-based	> \$500
Inventory Optimization	2 AM Sundays	None
Work Order Generation	Event-driven	None
Production Scheduling	5 AM daily	None
Order Fulfillment	Event-driven	None
Shipment Tracking	Every 2 hours	None
Supplier Performance Review	Monthly	None
Invoice Matching	Event-driven	None
Payment Processing	10 AM Mon/Wed/Fri	> \$2,000
Exception Handling	Threshold-based	Varies

Multi-Level Approval Hierarchy

The system enforces configurable approval thresholds to maintain financial controls:

Purchase Orders: - Auto-approve: up to \$1,000 - Level 1 (Ops Manager): \$1,001 -- \$5,000 - Level 2 (Admin): \$5,001 -- \$25,000 - Level 3 (Executive): \$25,000+

Payments: - Auto-approve: up to \$2,000 - Level 1 (Ops): \$2,001 -- \$10,000 - Level 2 (Admin): \$10,001 -- \$50,000 - Level 3 (Executive): \$50,000+

AI Decision Making

The AI agents make decisions across: - **Vendor selection** based on historical performance scoring - **Reorder quantity calculation** using demand forecasts and safety stock models - **Production scheduling** optimized for capacity and material availability - **Freight carrier selection** balancing cost vs. transit time - **Quote evaluation** with automatic accept/reject recommendations - **Anomaly detection** with automatic exception routing

Deployment & Infrastructure

Deployment Options

Platform	Method
Vercel	One-click deploy, automatic CI/CD from GitHub
Railway	Managed Node.js hosting with auto-detection
Self-hosted	Any Node.js 18+ environment; single <code>dist/index.js</code> bundle serves both API and static frontend

Production Build

The build produces two artifacts: 1. `dist/public/` -- Static frontend assets (served by Express) 2. `dist/index.js` -- Bundled backend server

This single-binary deployment model simplifies infrastructure and reduces operational overhead.

Database

- MySQL with Drizzle ORM for type-safe queries and migrations
 - 169 tables with full relational integrity
 - Migration management via `drizzle-kit`
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Key Metrics

Metric	Value
Total codebase	128,000+ lines of TypeScript
Database tables	169
Schema definition	5,076 lines
Frontend pages	90
UI components	69
Backend services	23+
Test suites	31
Integrations	7 (QuickBooks, Shopify, Google Workspace, SendGrid, Fireflies, IMAP, Airtable)
Autonomous workflows	14
AI assistant tools	20+
User role types	10
Supported auth methods	5 (email/password, Google, Apple, Microsoft, GitHub)

Competitive Positioning

vs. Legacy ERP (NetSuite, SAP Business One)

Dimension	Legacy ERP	AI ERP System
AI Integration	Bolt-on, limited	Native, autonomous agents
Implementation	6-18 months, \$100K+	Self-serve, deploy in minutes
Customization	Consultant-dependent	Code-level flexibility
Supply Chain Automation	Manual workflows	14 autonomous workflows with human-in-the-loop
Natural Language Interface	None	Full NLP command bar and conversational AI
Partner Portals	Separate add-ons	Built-in copacker, vendor, and supplier portals
Document Intelligence	Basic OCR add-ons	Vision-based OCR with automatic record creation
Cost	\$50K-500K+/year	Fraction of legacy pricing

vs. Modern ERPs (Cin7, Katana, Fishbowl)

Dimension	Modern Point Solutions	AI ERP System
Scope	1-3 modules, requires integrations	All-in-one: 10+ modules
AI Capabilities	Basic reporting	Autonomous agents, NLP, OCR, forecasting
CPG-Specific	Generic	Purpose-built for CPG with copacker management
Fundraising/CRM	Not included	Built-in investor CRM, cap tables, data rooms
Integration Complexity	Multiple tools to connect	Single platform, native integrations

Target Market

Primary ICP: CPG brands and manufacturers with \$2M--\$50M in revenue managing: - Multi-warehouse inventory operations - Copacker and contract manufacturer relationships - Complex

vendor/supplier networks - Production planning and BOM management - Omnichannel sales (DTC + wholesale + marketplace)

Secondary ICP: Growth-stage consumer brands actively fundraising, who benefit from the integrated CRM, cap table, and data room features alongside operational ERP.

Product Roadmap Highlights

Building on the current platform, planned enhancements include: - Advanced demand forecasting with external data signals (weather, seasonality, promotions) - EDI (Electronic Data Interchange) for enterprise retail compliance - Multi-currency and multi-entity support for international operations - Mobile-native companion app for warehouse and production floor - Expanded marketplace integrations (Amazon, Walmart, Target+) - Enhanced AI agent capabilities with reinforcement learning from approval feedback

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